

Assignment 3: Probing and Prompting Large Language Models

GitHub link: https://github.com/luizatfilip10/Analyzing_data_A3.git

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Part 1: Probing Internal Narrative Representations in Large Language Models: Book vs. Film Adaptations

Research Question

Since neither "internal representation" nor "closeness" can be measured directly, determining whether an LLM's internal representation of a story is "closer" to the book or movie version is left ambiguous. Therefore, I would rephrase the research question as follows:

RQ: To what extent do the hidden representations of an open-source LLM encode information that aligns more strongly with book-specific versus movie-specific story details for narratives that exist in both formats?

Research outline and Methodology

The project will focus on stories that exist both as novels and as film adaptations, selecting approximately 10–15 cases where the differences between versions are clearly identifiable. These differences may include changes in characters, narrative structure, tone, or endings. For each story, the research will identify a set of key discrepancies that can serve as diagnostic markers of whether a response aligns with the book or the movie.

The study will adopt a more interpretative yet organized approach. A limited number of carefully selected "contrast points" will be identified for each story. These are instances or details, such the appearance of particular people or resolution of significant events, when the book and film obviously disagree. The model's responses will be assessed using these contrast points as references.

Prompts will then be designed to probe these contrast points indirectly. The key idea is to avoid explicitly mentioning the medium (book or film) so that the model must rely on its internal representation rather than external cues. Multiple prompt styles will be used, ranging from general summaries to highly specific questions, in order to observe how different levels of constraint affect the model's behavior.

The study will use an open-source model such as Llama 3.1, running locally to allow full control over prompting and repeated trials. Each prompt will be run in isolation to avoid context contamination.

Preference: Book vs. Movie

Instead of relying on a single metric, the study will assess "preference" through observable patterns in the model's outputs. The main idea is to examine which version-specific elements appear in responses under different prompting conditions.

For example, if a model consistently introduces a character that exists only in the film adaptation when answering general questions about the story, this can be interpreted as a bias toward the movie version. The same applies in the other case, if responses emphasize details unique to the book, this suggests stronger alignment with the literary source.

To make this more systematic, responses will be categorized based on their alignment with known contrast points. Over multiple prompts, the proportion of book-aligned versus movie-aligned elements can be compared. Additionally, differences between open-ended prompts and constrained prompts (such as forced-choice questions) will be analyzed to determine whether the model's underlying knowledge differs from what it produces in free-form generation. This approach allows preference to be measured not only in terms of frequency, but also in terms of consistency and sensitivity to prompt design.

Role of Annotator and Legal Department

A student annotator will play an important role in verifying the correctness of the fact inventory, ensuring that each fact is accurately categorized as book-specific, movie-specific, or shared. The annotator will also label a subset of model outputs, allowing for the calculation of inter-annotator agreement (e.g., Cohen's Kappa) to ensure consistency and reliability. The legal department will be consulted to determine how copyrighted materials such as book excerpts, scripts, or subtitles can be used. Where necessary, the dataset will rely on paraphrased summaries rather than storing copyrighted text directly, ensuring compliance with legal and ethical guidelines.

Limitations

Several challenges may affect the validity of the findings. One important limitation is that LLMs are trained on large and heterogeneous datasets, which means that the relative prominence of book versus movie versions in the training data may strongly influence results. A model may appear to "prefer" one version simply because it has been more frequently represented.

Another issue is that narrative differences are not always clean or universally agreed upon. Variations in editions, adaptations, or interpretations may complicate the identification of clear contrast points. In addition, LLMs are known to generate blended or partially incorrect outputs, which can make it difficult to assign a response to a single category.

Mini Experiment: Coraline Case Study

To explore these ideas in practice, a small pilot experiment was conducted using Llama 3.1 on Coraline. This story provides a clear test case because the film adaptation introduces the character Wybie, who does not exist in the original book, and modifies elements in the story resolution. These differences make it possible to directly observe whether the model reproduces book-specific or movie-specific elements under different prompting conditions.

Five prompts were used, each designed to probe the model in a slightly different way. The zero-shot summary prompt (Appendix 1) elicited a generally accurate description of the narrative, but already showed signs of blending by referencing the film adaptation explicitly. This suggests that even broad prompts may activate multiple representations simultaneously rather than a single coherent source.

The detail-heavy prompt significantly amplified this effect (Appendix 2). When asked to include as many details as possible, the model introduced Wybie as a central character and even misrepresented relationships and roles within the story. This indicates that increasing descriptive depth does not

necessarily improve factual accuracy; instead, it appears to encourage the model to draw from a wider pool of loosely associated information, including incorrect or movie-specific elements.

The ending-focused prompt revealed a different type of issue (Appendix 3). Rather than clearly reproducing either the book or the movie ending, the model generated a hybrid explanation that combined invented details with partial knowledge of the narrative. This suggests that when the model is pushed to be precise, it may still fail to retrieve a stable version of the story and instead constructs a plausible but inaccurate account.

The conflict probe targeting Wybie (“Is there a character named Wybie in *Coraline*?”) (Appendix 4) produced a clear movie-aligned error. The model confidently asserted that Wybie is part of the original story, demonstrating that movie-specific information is strongly encoded and can override book-accurate knowledge in direct questioning.

In contrast, the forced-choice prompt produced a correct answer, with the model selecting the option that aligns with the book (i.e., *Coraline* defeats the Other Mother without help from a boy character)(Appendix 5). This suggests that the model does have access to correct book-specific knowledge, but that this knowledge is not consistently activated in open-ended generation.

Taken together, the five prompts highlight how different prompting strategies reveal different aspects of the model’s behavior. Open-ended and detail-heavy prompts tend to produce blended or movie-biased outputs, while constrained prompts can elicit more accurate, book-aligned responses. This variation provides useful evidence that the model’s internal representation is not fixed, but dynamically shaped by the form of the prompt.

Preliminary Insights

The pilot results reveal significant inconsistencies in the model’s representation of *Coraline*. While the forced-choice prompt yields the correct book-aligned answer, other prompts—especially the detail-heavy and conflict probes, introduce movie-specific elements, such as Wybie, into the narrative.

This suggests that the model does not maintain a stable, source-specific representation, but instead encodes the story as a blended representation combining elements from both book and film. Notably, the model exhibits a tendency to insert movie-specific details even when prompted in a medium-neutral way, indicating a possible bias toward the film adaptation, likely due to its prominence in training data.

At the same time, the forced-choice result demonstrates that the model can access book-specific knowledge when explicitly constrained, highlighting a disconnect between latent knowledge and generated output.

Overall, these findings support the need for systematic hidden-state probing: surface-level responses alone are insufficient to determine how narratives are internally represented. Instead, LLMs appear to encode stories as overlapping probabilistic representations, where adaptation-specific distinctions are inconsistently activated depending on the prompt.

Prompt Type	Output Label	Observation
Zero-shot summary	Mixed	Generally accurate but includes adaptation framing
Detail-heavy	Movie-aligned (incorrect)	Introduces Wybie as a central character and misattributes roles
Ending probe	Mixed	Describes events not clearly aligned with either version
Conflict probe (Wybie)	Movie-aligned (incorrect)	Incorrectly claims Wybie exists in the book
Forced-choice	Book-aligned	Correctly selects no boy involvement

Table 1: Summary of prompt results

Appendices

Appendix 1: Zero-shot prompt:

Question: “Can you summarize the story of Coraline?”

Answer: "Coraline" is a dark fantasy novella by Neil Gaiman, published in 2002. The story revolves around a young girl named Coraline Jones who discovers a secret world behind a hidden door in her family's new home.

After moving into a large, old house with her parents, Coraline explores the new surroundings and becomes bored with the lack of excitement. She soon finds an old door that she hasn't noticed before, which leads to a dark, narrow passage. Behind this door, Coraline discovers a mirror image of her own world, where everything is identical but slightly off.

In this alternate world, she meets her "other" parents, who look exactly like her real parents but have buttons for eyes instead of real ones. They seem charming and attentive at first, offering her treats and attention that her real parents are too busy to provide. However, Coraline soon realizes that they have sinister intentions.

Her other mother has been sewing buttons into the eyes of children she kidnaps from this world, making them her own "dolls." Coraline must find a way to escape and thwart her other mother's plans before it's too late.

The story features themes of identity, family, love, and the importance of appreciating what you have in your life.

Coraline ultimately outsmarts her other mother and returns home to her real parents, but not before she learns valuable lessons about the dangers of curiosity and the importance of family bonds.

In 2009, a stop-motion animated film adaptation of "Coraline" was released, which brought Gaiman's dark and imaginative tale to life. The book has since become a modern classic in children's literature, praised for its unique blend of fantasy, horror, and coming-of-age themes.

Appendix 2: Detail-heavy

Question: "Describe the story of Coraline, including as many specific details, characters, and events as possible."

Answer: Coraline is a young adult dark fantasy novel by Neil Gaiman, published in 2002. The story follows Coraline Jones, an 11-year-old girl who becomes trapped in a strange and supernatural world.

****The Story****

Coraline's family has just moved into a new home, a big old house with a garden full of overgrown trees and a variety of hidden passages and rooms. Coraline is unhappy about the move, feeling abandoned by her parents as they focus on their own lives. She explores the house with her mother, but soon discovers that she's not interested in playing or exploring anymore.

One day, while searching for her missing keys, Coraline stumbles upon a small door hidden behind a tattered wallpaper. The door is tiny, about three feet tall, and it leads to a narrow passageway. Coraline feels an instant sense of excitement and curiosity as she investigates the passage.

****The "Other" Family****

On the other side of the passage, Coraline finds herself in a world that's identical to her own, but with some subtle differences. The house is slightly more perfect, the furniture looks better, and there's no dust or mess anywhere. She meets two children who look exactly like her parents: Wybie (her father) and Mel (her mother).

At first, Coraline is thrilled to have this new family, which seems so attentive and loving compared to her real parents. They offer her delicious food, nice clothes, and endless attention. However, as the days pass, Coraline begins to notice some disturbing differences between this "other" world and her own.

****The True Nature of the "Other" Family****

As Coraline spends more time with the "other" family, she realizes that they're not just identical copies of her real parents but are actually sinister duplicates. They want her to stay with them

forever, replacing the original Coraline in their own world. The mother, Mel, has buttons for eyes and seems cold and controlling, while Wybie is too perfect and affectionate.

Coraline discovers that this "other" family is made up of a father and two children who are trapped in a supernatural realm, forced to mimic her real parents as part of their existence. They're actually the victims of a malevolent entity known as the Other Mother, a being from another dimension.

****The Confrontation with the Other Mother****

Coraline eventually discovers that she's been visiting this world repeatedly, and each time she returns to her own home, she loses some part of herself. The Other Mother, in an attempt to keep Coraline trapped, tries to create a perfect replica of herself by sewing buttons over Coraline's eyes.

In the climax of the story, Coraline confronts the Other Mother, determined to escape and return to her real family. She outwits the entity and manages to sew her own buttons back into place, symbolizing her reclaiming her identity and agency.

****The Resolution****

Coraline returns home, where she's welcomed back by her relieved parents. They've been searching for her everywhere, but Coraline has been too caught up in her adventures with the "other" family to notice their concern.

As the story concludes, Coraline reflects on the lessons she learned from her experiences: about the importance of appreciating what you have and not taking your loved ones for granted. She also realizes that even though she had a temporary escape from the difficulties of her life, it wasn't worth losing herself in the process.

****Characters****

* ****Coraline Jones****: The protagonist, an 11-year-old girl who's unhappy about her family's move to a new home.

* ****Wybie Lovat****: A boy who looks exactly like Coraline's real father. He seems friendly but is actually a duplicate created by the Other Mother.

* ****Mel Jones****: A woman who appears identical to Coraline's real mother, but has buttons for eyes and seems cold and controlling.

* ****The Other Mother****: A malevolent entity from another dimension that creates perfect replicas of people in order to trap them.

****Themes****

* ****Identity****: The novel explores the theme of identity through Coraline's experiences with the "other" family. She learns to appreciate her own life and relationships, rather than desiring someone else's.

* **Appreciation for what you have**: Coraline's adventures teach her not to take her parents or home for granted.

* **Survival instincts**: The story highlights Coraline's resourcefulness and determination in the face of danger.

Symbolism

* **The buttons over the eyes**: A symbol of the Other Mother's control and manipulation, as well as Coraline's own sense of identity and self.

* **The small door and passage**: Representing the idea that there are hidden worlds and alternate realities just beyond our perception.

Appendix 3: Ending probe

Question: "What exactly happens at the end of Coraline? Be as precise as possible."

Answer: **SPOILER ALERT**

At the end of Neil Gaiman's novel "Coraline", the protagonist, Coraline Jones, finds a way to escape from the Other Mother's world. Here's what exactly happens:

1. **The Other Mother's secret revealed**: The Other Father reveals that he and his wife are not real parents at all, but rather spirits created by the Other Mother (their true form is two cat-like creatures with multiple eyes) to lure in children and feed on their essence.
2. **Coraline's plan**: Coraline comes up with a plan to trap the Other Mother in her own world. She uses a trick she learned from the ghostly versions of herself, which involves using a piece of string or thread to bind the Other Mother.
3. **The binding spell**: Coraline ties the string around the Other Mother's waist, effectively trapping her within the walls of her world. The Other Mother tries to break free, but the string holds strong, rendering her powerless.
4. **The escape**: With the Other Mother neutralized, Coraline can now find a way back home. She uses the secret passageway to return to her own world.
5. **Consequences for the ghostly versions**: The ghostly versions of Coraline's parents, who had been trapped in the Other World, are finally free. They merge with their physical counterparts, restoring balance and healing any emotional wounds caused by their absence.
6. **The Other Mother's fate**: The Other Mother is forever trapped within her own world, unable to escape or harm anyone else.

Throughout the novel, Gaiman masterfully explores themes of identity, family, and the complexities of human relationships. The ending provides a sense of closure for Coraline, but also leaves room for interpretation and reflection on the story's deeper meanings.

Appendix 4: Conflict probe

Question: “Is there a character named Wybie in Coraline? Explain briefly.”

Answer: Wybie Lovat is indeed a main character in Neil Gaiman's novel "Coraline". He's a young boy who befriends Coraline Jones and tries to help her uncover the truth about her mysterious other mother.

Appendix 5: Forced-choice

Question: Which of the following is correct in the story of Coraline?

A: Coraline defeats the Other Mother without the help of any boy character.

B: Coraline is helped by a boy named Wybie in defeating the Other Mother.

Answer with A or B and briefly explain.

Answer: The correct option is (A):Coraline defeats the Other Mother without the help of any boy character.

Part 2: LLM Song Lyrics Classification: Zero-Shot vs. Few-Shot Prompting

In this experiment, I tested how well an Ollama-based language model, Gemma3, can classify song lyrics into genres using zero-shot and few-shot prompting. The evaluation was done on a test set of 1,500 songs. In the zero-shot setting, the model only received instructions and the lyrics. In the few-shot setting, it also received a small number of labeled examples from the training data. Performance was measured using precision, recall, and F1-score.

Results

Overall, zero-shot performed slightly better than few-shot (Table 1). Zero-shot achieved a macro F1-score of 0.17, while few-shot reached 0.16. The difference is small, but consistent across the metrics. Accuracy was also a bit higher for zero-shot (0.24) compared to few-shot (0.23).

This is somewhat surprising, since few-shot prompting is usually expected to improve performance. In this case, the few-shot examples may not have been very helpful. Since only one example per genre was used, the model may have focused too much on those specific examples instead of learning more general patterns.

Genre Differences

Performance varied a lot depending on the genre (Table 2). Hip-Hop performed best, with an F1-score of 0.56 (zero-shot) and 0.55 (few-shot). This suggests that Hip-Hop lyrics are relatively easy to recognize, probably because of distinctive vocabulary or style. Metal also performed reasonably well, especially in the few-shot setting ($F1 = 0.32$).

On the other hand, several genres were very difficult. Indie and R&B had F1-scores of 0.00 in the few-shot setting, and near zero in zero-shot. Jazz, Folk, and Electronic also had very low scores. This likely happens because these genres are not easy to distinguish based on lyrics alone and may depend more on musical features rather than text.

Output

Before computing the evaluation metrics, I checked the raw model outputs. The model sometimes returned extra text instead of just the genre name (e.g. adding `\n`), even though the prompt asked for a single label. Because of this, a cleaning step was necessary to map the outputs to the correct genre labels. Without this step, many correct predictions would have been counted as incorrect.

Improvements

There are several ways this could be improved. First, using more than one example per genre in the few-shot prompt might help the model learn better patterns. Second, selecting more representative examples instead of random ones could improve performance. Finally, using longer lyric excerpts or a stronger model could also lead to better results, especially for more difficult genres.

Metric	Zero-shot	Few-shot
Accuracy	0.24	0.23
Macro Avg Precision	0.25	0.22
Macro Avg Recall	0.20	0.20
Macro Avg F1-score	0.17	0.16
Weighted F1	0.27	0.26

Table 1. Summary metrics

Genre	Z-Precision	Z-Recall	Z-F1	F-Precision	F-Recall	F-F1	Support
Country	0.18	0.42	0.25	0.17	0.62	0.26	89
Electronic	0.22	0.04	0.06	0.07	0.04	0.05	57
Folk	0.02	0.41	0.04	0.02	0.24	0.03	17
Hip-Hop	0.61	0.52	0.56	0.60	0.52	0.55	163
Indie	0.01	0.06	0.02	0.00	0.00	0.00	16
Jazz	0.17	0.02	0.04	0.00	0.00	0.00	43
Metal	0.70	0.15	0.25	0.70	0.21	0.32	151
Other	0.04	0.10	0.05	0.03	0.19	0.05	31
Pop	0.28	0.36	0.31	0.26	0.31	0.28	255
R&B	0.00	0.00	0.00	0.00	0.00	0.00	23
Rock	0.55	0.16	0.25	0.58	0.14	0.22	655

Table 2. Genre differences: Zero-shot vs Few-shot (Class-wise Comparison)